C1.2 Protocol overview

This section introduces the ACE protocol.

C1.2.1 About the ACE protocol

The ACE protocol extends the AXI4 protocol and provides support for hardware-coherent caches. The ACE protocol is realized using:

- A five state cache model to define the state of any cache line in the coherent system. The cache line state determines what actions are required during access to that cache line.
- Additional signaling on the existing AXI4 channels that enables new transactions and information to be conveyed to locations that require hardware coherency support.
- Additional channels that enable communication with a cached master when another master is accessing an address location that might be shared.

The ACE protocol also provides:

- Barrier transactions that guarantee transaction ordering within a system, see *Barriers* on page C1-148
- *Distributed Virtual Memory* (DVM) functionality to manage virtual memory, see *Distributed Virtual Memory* on page C1-149.

C1.2.2 Coherency model

Figure C1-1 shows an example coherent system that includes three master components, each with a local cache. The ACE protocol permits cached copies of the same memory location to reside in the local cache of one or more master components.

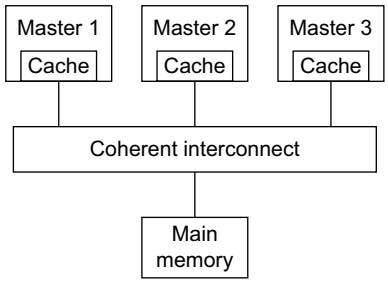


Figure C1-1 Example coherent system

The ACE coherency protocol ensures that all masters observe the correct data value at any given address location by enforcing that only one copy exists whenever a store occurs to the location. After each store to a location, other masters can obtain a new copy of the data for their own local cache, allowing multiple copies to exist.

A cache line is defined as a cached copy of a number of sequentially byte addressed memory locations, with the first address being aligned to the total size of the cache line.

There is no requirement to keep main memory up to date at all times. Main memory is only required to be updated before a copy of the memory location is no longer held in any shareable cache.

Note

Although not a requirement, it is acceptable to update main memory while cached copies still exist.

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The ACE protocol enables master components to determine whether a cache line is the only copy of a particular memory location, or if there might be other copies of the same location, so that:

- if a cache line is the only copy, a master component can change the value of the cache line without notifying any other master components in the system
- if a cache line might also be present in another cache, a master component must notify the other caches, using an appropriate transaction.

C1.2.3 Cache state model

To determine whether an action is required when a component accesses a cache line, the ACE protocol defines cache states. Each cache state is based on a cache line characteristic.

The cache line characteristics are:

Valid, Invalid When valid, the cache line is present in the cache. When invalid, the cache line is not present in the cache.

Unique, Shared When unique, the cache line exists only in one cache. When shared, the cache line might exist in more than one cache, but this is not guaranteed.

Clean, Dirty When clean, the cache does not have responsibility for updating main memory. When dirty, the cache line has been modified with respect to main memory, and this cache must ensure that main memory is eventually updated.

Figure C1-2 shows the ACE five state cache model and Table C1-1 on page C1-136 provides more information about each state.

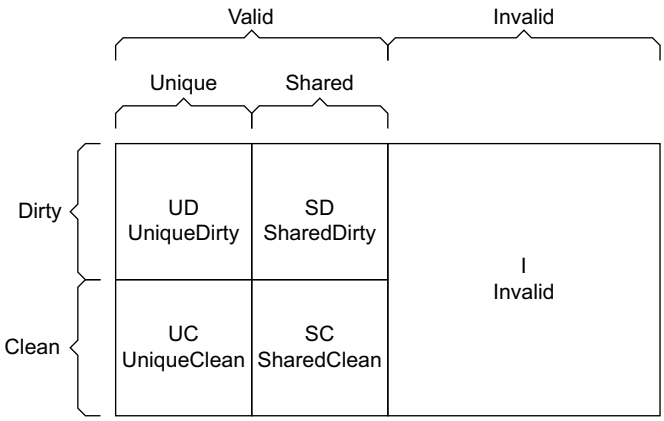


Figure C1-2 ACE cache state model

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Table C1-1 ACE cache states

State	Abbreviation	Description
Invalid	I	The cache line does not exist in this cache.
UniqueClean	UC	The following rules apply to a cache line that is in the UniqueClean state: • the cache line is held only in this cache and it has not been modified with respect to main memory • a component can perform a store to the cache line without notifying other caches.
UniqueDirty	UD	The following rules apply to a cache line that is in the UniqueDirty state: • the cache line is held only in this cache • the cache line has been modified with respect to main memory and this cache must ensure that the changes are subsequently notified to main memory • a component can perform subsequent stores to the cache line without notifying other caches.
SharedClean	SC	The following rules apply to a cache line that is in the SharedClean state: • the cache line might be shared with another cache • it is not known if the cache line is modified with respect to main memory, but this component is not responsible for updating main memory • a component must notify other caches before performing a store to the cache line.
SharedDirty	SD	The following rules apply to a cache line that is in the SharedDirty state: • the cache line might be shared with another cache • the cache line has been modified with respect to main memory and this cache must ensure that the changes are subsequently notified to main memory • a component must notify other caches before performing a store to the cache line.

Cache state rules

The rules that apply to the cache states are:

- A line in a Unique state must only be in one cache.
- A line that is in more than one cache must be in a Shared state in every cache it is in.
- When a cache obtains a new copy of a line, other caches that also have a copy of the line, and might have the line in a Unique state, must be notified to hold the line in a Shared state.
- When a cache discards a copy of a line, there is no requirement to inform other caches that also have a copy of the line. This means that a line in a Shared state might be held in only one cache.
- A line that has been updated, relative to main memory, must be in a Dirty state in one cache.
- A line that has been updated, relative to main memory, and is in more than one cache, must be in a Dirty state in only one cache.

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